WEEK 5

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CODE :

```
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main()
{
    int p[2];
    char msg[] = "Hello";
    char buffer[20];
pipe(p);

if(fork() == 0)
{
    close(p[1]);

    read(p[0], buffer, sizeof(buffer));

    printf("Child received: %s\n", buffer);
}
else
{
    close(p[0]);

    write(p[1], msg, sizeof(msg));

    printf("Parent sent: %s\n", msg);
}

return 0;
}
```

```
princy@Samuel:~$ nano week5_ProducerConsumer.c
princy@Samuel:~$ gcc week5_ProducerConsumer.c
princy@Samuel:~$ ./a.out
Parent sent: Hello
Child received: Hello
princy@Samuel:~$
```

```
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    int p[2];
pipe(p);

if(fork() == 0)
{
    close(p[0]);

    dup2(p[1], 1);

    execlp("ls", "ls", "-l", NULL);
}

if(fork() == 0)
{
    close(p[1]);

    dup2(p[0], 0);

    execlp("grep", "grep", ".c", NULL);
}

close(p[0]);
close(p[1]);

wait(NULL);
wait(NULL);

return 0;
}
```

```
princy@Samuel:~$ nano week5.c
princy@Samuel:~$ gcc week5.c
princy@Samuel:~$ ./a.out
-rw-r--r-- 1 princy princy    15 Jul 29 11:30 1.txt
-rw-r--r-- 1 princy princy    51 Jul 29 11:32 3.txt
-rw-r--r-- 1 princy princy   691 Aug 22 09:21 W2.c
-rwxr-xr-x 1 princy princy 16208 Nov  1 05:55 a.out
-rw-r--r-- 1 princy princy   185 Aug  4 05:56 f2.c
drwxr-xr-x 2 princy princy  4096 Jul 29 11:35 hello.c
-rw-r--r-- 1 princy princy   327 Aug  4 04:54 l2.c
-rw-r--r-- 1 princy princy   567 Aug  4 05:45 l3.c
-rwxr-xr-x 1 princy princy 16688 Sep 30 16:39 myshell
-rw-r--r-- 1 princy princy  2085 Sep 30 16:38 myshell.c
-rw------- 1 princy princy    97 Aug 25 07:01 open.c.save
-rw-r--r-- 1 princy princy   204 Aug  1 09:47 p2.c
-rw-r--r-- 1 princy princy   175 Aug  1 09:59 pid.c
-rw-r--r-- 1 princy princy   155 Aug  1 09:25 process2.c
-rw-r--r-- 1 princy princy   240 Aug 25 06:04 read.c
-rw-r--r-- 1 princy princy   328 Aug 25 06:07 read1.c
drwxr-xr-x 3 princy princy  4096 Oct  6 05:24 system-call-monitor
-rw-r--r-- 1 princy princy   779 Aug 17 06:41 w1.c
-rw-r--r-- 1 princy princy   640 Aug 17 08:29 w2.c
-rw-r--r-- 1 princy princy   992 Aug 22 09:24 w3.c
-rw-r--r-- 1 princy princy   990 Aug 22 09:29 w4.c
-rw-r--r-- 1 princy princy   326 Nov  1 04:54 week10.c
-rw-r--r-- 1 princy princy   431 Nov  1 05:54 week5.c
-rw-r--r-- 1 princy princy   490 Aug 22 09:36 zombie.c
princy@Samuel:~$
```